

Arjun Mauji

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Professional Summary

Mechanical Engineering undergraduate at Queen's University with hands-on experience in mechanical and embedded systems design, along with strong problem-solving and communication skills developed through collaborative engineering projects.

Education

B.A.Sc. Mechanical Engineering, Queen's University

Kingston, Canada | 3rd/4

- Dean's Scholar 24/25, Biggs – Ronald & Deanna Scholarship
- Coursework/Concepts: Embedded Programming, Real-time Operating Systems, Microcontrollers, Numerical Methods & Statistics, Heat Transfer, Fluid Mechanics

Work Experience

Teaching Assistant (Mechatronics Course), Queen's University

Sep – Dec 2025

- Co-supervised 10 mechatronics labs with over 100 students on measurement sensors and signal noise filtering
- Answered electronic measurement theory questions and troubleshoot lab circuits

Software Developer, Queen's EngSociety Software Team

Feb – March 2025

- Developed a Raspberry Pi based interactive display providing campus information to 200+ students daily.
- Maintained webpage response time through low latency event handling of IR touchframe input in C
- Implemented Vercel CD workflow to automate code deployments

Projects

Embedded Software Team, Queen's Off-Road Car Design Team (Baja)

Feb 2026 – Present

- Migrated the existing CAN protocol library to use FreeRTOS, improving real-time sensor data-logging reliability
- Programmed STM32 boards in C to monitor the thermal condition of the transmission system's metal components

Mechanical Team, Queen's Off-Road Car Design Team (Baja)

Sep – Dec 2024

- Assisted in the remodelling and manufacturing of the continuously variable transmission (CVT) using SolidWorks
- Reduced the primary transmission's component's machining time by over 30 minutes through weight reduction

Hardware Motion Analyzer, Personal Biomedical Project

Nov – Dec 2025

- Prototyped a wireless motion analyzer to test the accuracy of a portable biomechanics tool under \$40
- Programmed three ESP32-based sensor modules with motion sensors to capture & evaluate multi-joint angles
- Implemented RTOS tasking with I2C communication to log and filter up to 20 angle data streams in parallel
- Quantified the motion through analysis of movement consistency and peak angle values with SciPy

Skills

Mechanical: SolidWorks CAD & FEA, Assembly Drawings (GD&T), FMEA

Embedded Systems: STM32 (ARM Cortex-M), FreeRTOS, CAN, I2C

Software Tools: C, C++, Python (NumPy, Pandas, Matplotlib), MATLAB (Simulink), Git/GitHub, VS Code

Electrical Tools: Oscilloscopes, Multimeters, Soldering

Spoken Languages: Bilingual (Fluent) – English & French

Professional Tools: MS Outlook, Word, Excel, PowerPoint